

# AI Candidate Ranking System

E2E Pitch Deck & Evaluation Report

# Problem Statement

- Traditional ATS filters rely on simple keyword matching and are easily gamed.
- Candidates engage in keyword-stuffing (e.g. listing Python/AI skills with no experience).
- Mismatches exist between candidate titles and actual career description narratives.
- Scalability issues: processing 100,000+ profiles manually or with heavy LLMs is impossible.

# System Architecture

1. Data Ingestion: Stream JSONL to handle large 487MB dataset in constant memory.
2. Fast Semantic Filter: Compute lightweight Jaccard/Sentence Embeddings on all candidates.
3. Deep Scorer: Calculate precise skill fit weights, verification penalties & behavioral signals.
4. LLM Evaluator: Batch calls to DeepSeek API (with rate limiters) for narrative reasonings.
5. Fallback Mode: Seamless transition to rule-based templates if DeepSeek is offline.

# Scoring Methodology

Final Score =  $0.40 * \text{Technical} + 0.35 * \text{Career} + 0.25 * \text{Behavioral}$

- Technical (40%): Semantic similarity, required skill overlap, core tools.
- Career (35%): Experience (target 5-9, peak 6-8), current title, product company history.
- Behavioral (25%): Login recency, open to work status, response rate, notice period.
- Hard Filters & Honeypots: Auto 0.0 score (consulting-only, academic, impossible YoE).

# Key Data Insights

- Total candidates processed: 100
- Average candidate composite score: 0.4561
- Highly endorsed skills: Python, PyTorch, SQL.
- Notice period penalty successfully applied to long notice profiles.

# Top 5 Ranked Candidates

| Candidate ID | Composite Score | Reasoning Summary                                                 |
|--------------|-----------------|-------------------------------------------------------------------|
| CAND_0008295 | 0.5846          | A solid candidate with a background as a AI Research Engineer ... |
| CAND_0054859 | 0.5805          | Demonstrating exceptional career fit, this Senior Software Eng... |
| CAND_0010149 | 0.5676          | Demonstrating exceptional career fit, this ML Engineer possess... |
| CAND_0088025 | 0.5646          | This Staff Machine Learning Engineer brings 8.6 years of exper... |
| CAND_0080766 | 0.5475          | We highlight this profile due to strong career matching. The c... |